AP Calc Lesson 2.9 Homework

Answer Key

1. C, G, H

2. a. $f(x) = \csc x + 1$

   b. $f'(x) = -\csc x \cot x$

   c. $y - 3 = -2\sqrt{3}\left(x - \frac{\pi}{6}\right)$

3. a. $f(x) = \frac{1}{\cot x} = \tan x \Rightarrow f'(x) = \sec^2 x$

   b. $f(x) = \frac{x}{\cot x} \Rightarrow f'(x) = \frac{\cot x - x(-\csc^2 x)}{\cot^2 x} = \frac{\cot x + x\csc^2 x}{\cot^2 x} = \tan x + x\sec^2 x$

   c. $f(x) = \frac{\csc x}{\cot x} = \sec x \Rightarrow f'(x) = \sec x \tan x$

4. a. $\frac{ds}{d\theta} = -10\csc^2\theta$

   b. For $0 < \theta < \frac{\pi}{2}$, s decreases as θ increases.

   c. $s'(1) = -14.123$ meters per radian (or -14.122)

5. a. $g'(x) = f(x) \cdot -\csc^2 x + \cot x \cdot f'(x); g'\left(\frac{\pi}{6}\right) = f\left(\frac{\pi}{6}\right) \cdot \left(-\csc^2\left(\frac{\pi}{6}\right)\right) + \cot\left(\frac{\pi}{6}\right) \cdot f'\left(\frac{\pi}{6}\right) = -4 + 4\sqrt{3}$

   b. $h'(x) = \frac{(\sec x - 1)f'(x) - f(x)\sec x \tan x}{(\sec x - 1)^2}; h'(\pi) = \frac{(\sec \pi - 1)f'(\pi) - f(\pi)\sec \pi \tan \pi}{(\sec \pi - 1)^2} = -\frac{3}{4}$

6. a. Since $g'(x) = 1 + \sec^2 x$, which is greater than 0 for all x in the domain of g', $g'(x) \neq 0$ and a line tangent to g cannot be horizontal.

   b. $g'\left(\frac{5\pi}{4}\right) = 3$

   c. $x = \pi$

7. Since the slope of $Ax - By = 10$ is $\frac{A}{B}$, this value will provide the solution to our question. The slope of the line tangent to the curve must be equal to the slope of $Ax - By = 10$.
   Since $y' = 2\sec x \tan x, y'\Big|_{x=\frac{\pi}{6}} = 3\sec\left(\frac{\pi}{6}\right)\tan\left(\frac{\pi}{6}\right) = 3\left(\frac{2}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right) = 2$
   and $\frac{A}{B} = 2$. Thus, A is two times greater than B.

8. $\frac{\cot k - 0}{k - 2} = -\csc^2 k$
   $k = 2.484$ (or 2.483)

9. D

10. A